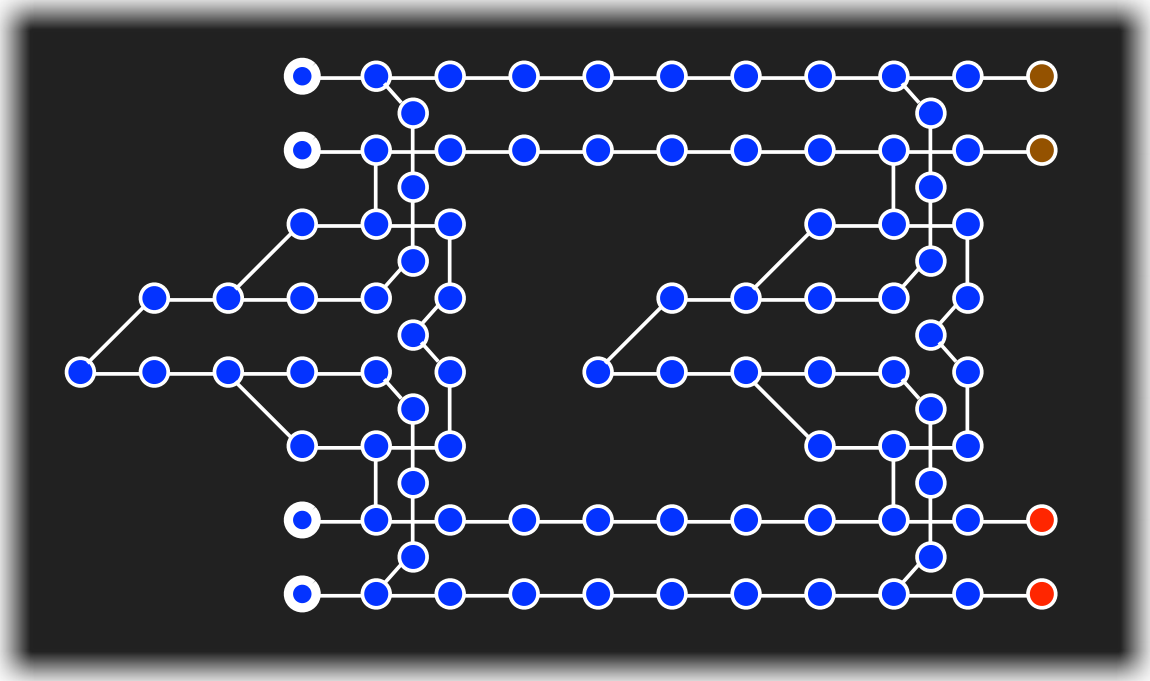
# Turning Stabilizer Codes Local

Dave Bacon and Jonathan Shi, University of Washington

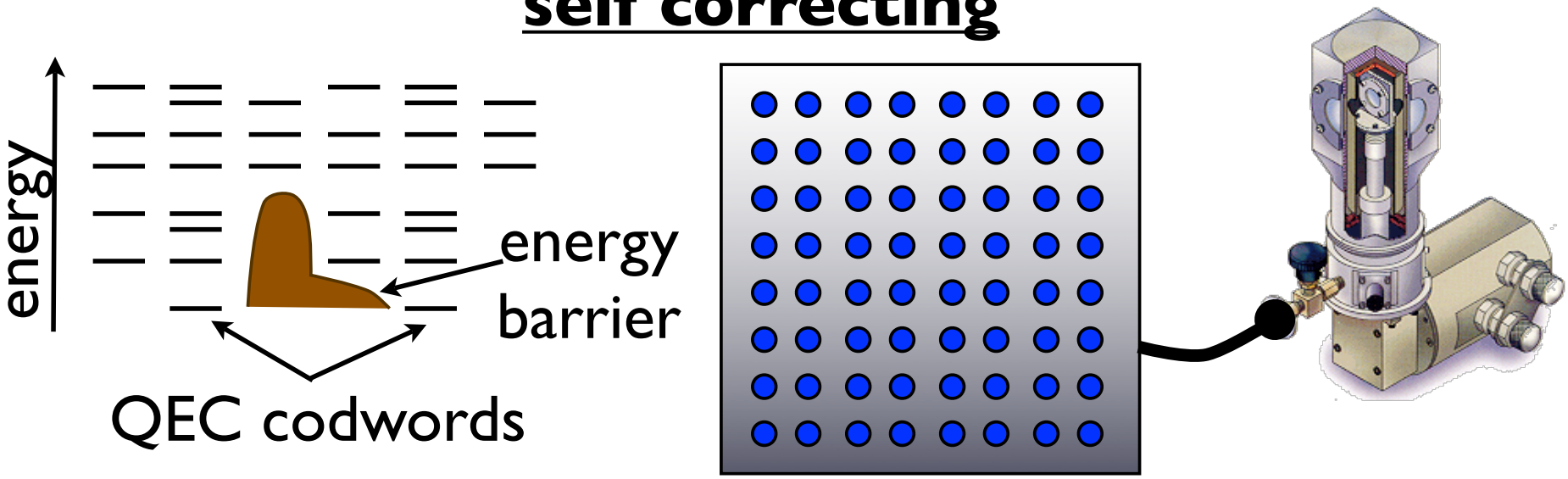


# A QuEST

Do self-correcting quantum systems exist in realistic settings?

realistic = local in less than three spatial dimension
+ few-body interactions

**self correcting**




quantum error correction is performed by simply keeping the many-body system cold

# Not Realistic (But Cool)

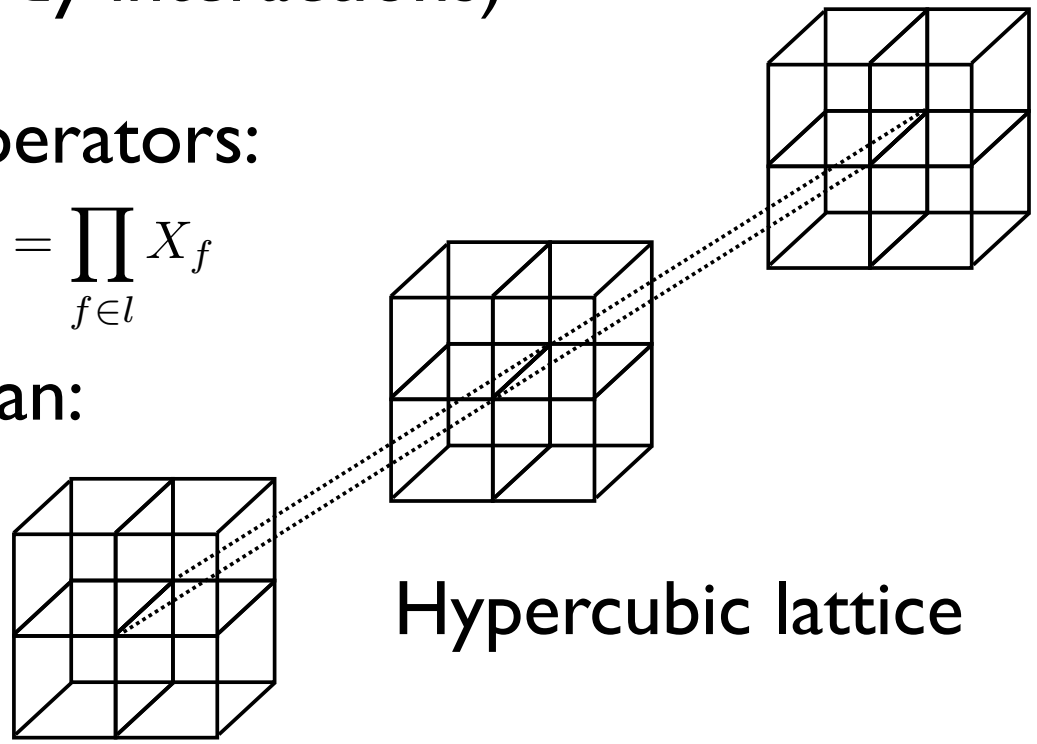
The toric code in 4 spatial dimensions is a local self-correcting system (six body interactions)

cube operators: link operators:

$$S_c = \prod_{f \in c} Z_f \qquad S_l = \prod_{f \in l} X_f$$

4D toric code Hamiltonian:

$$H = -J \left[ \sum_{c \in C} S_c + \sum_{l \in L} S_l \right]$$



Hypercubic lattice

Self correction comes from excitations being string-like (require energy prop to area)

# Codes

4D toric code has

a) Local check operators

b) Distance (lowest weight error that is not detectable) that scales as $L^2$ on 4D lattice of length L

2D toric code and 2D color codes have

a) Local check operators

b) Distance (lowest weight error that is not detectable) that scales as L on 4D lattice of length L

Most interesting QEC codes

a) Do not have local check operators

b) Can have distance that scales as number of qubits

Can we fix a)?

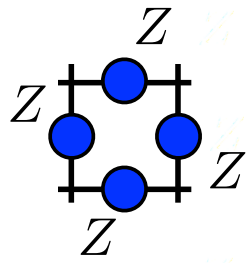
# Definitions

weight of Pauli operator = # of operators ≠ I

I⊗X⊗X⊗Y⊗Y⊗I⊗Z⊗I has weight 5

c-Local code: exists a set of stabilizer generators whose weight ≤ c

Ex: toric code is 4-local




c-Neighboring code: (in D dimensions): exists a set of stabilizer generators where all ≠ I operators are within a distance c of each other on some lattice of qubits in D spatial dimensions

Ex: surface code is 2-neighboring in 2D

# Main Result

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] stabilizer subsystem code, $N=O(n^2)$, whose syndrome measurements can be made using c-neighboring measurements (c constant) in $D \geq 2$ dimensions.

Translation: every stabilizer code can transformed into a local code at the cost of using more qubits!

# Intuition of the Construction

1. While stabilizers for a QEC code are not local, the QEC syndrome measuring circuits can be made local. QEC is local in space-time.

2. Measurement-based quantum computing takes a quantum circuit that is local in space-time and shows how, on a lattice where time has been converted to a space dimension, to
   a) locally specify a state
   b) enact the quantum computation by locally measuring this state

Space-time local QEC + MBQC = Spatially local codes?

# There's a Code in My Wire



Vertex operators at v: $X_v \prod_{(w,v)\in E} Z_w$

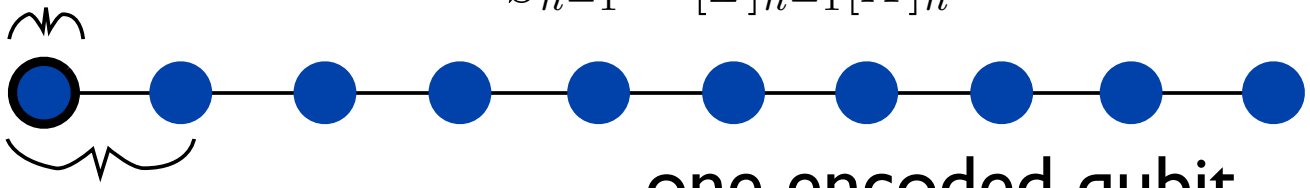
Consider stabilizer code with stabilizer generators all vertex operator on a line except first one:

n-1 stabilizer generators

$$S_i = [Z]_i[X]_{i+1}[Z]_{i+2} \quad 2 \le i \le n-2$$

$$S_{n-1} = [Z]_{n-1}[X]_n$$

Z info localized here

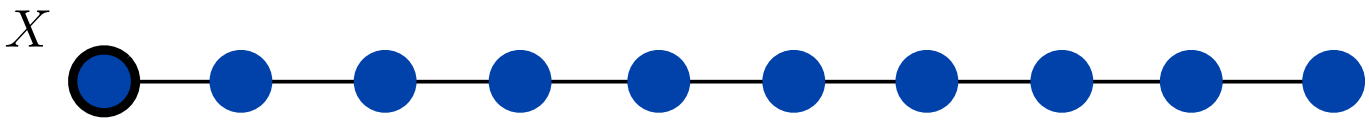


X info localized here

one encoded qubit

$$\bar{Z} = [Z]_1 \quad \bar{X} = [X]_1[Z]_2$$

# There's a Code in My Wire




Add a gauge qubit (X on first qubit)

new stabilizer: $S_i = [Z]_i[X]_{i+1}[Z]_{i+2} \quad 3 \leq i \leq n-2$
$S_{n-1} = [Z]_{n-1}[X]_n$

new gauge logical operators: $\tilde{X}_1 = [X]_1 \quad \tilde{Z}_1 = [Z]_1[X]_2[Z]_3$

old logical operators

$\bar{X} = [X]_1[Z]_2$
$\bar{Z} = [Z]_1$

no longer commute with gauge operators

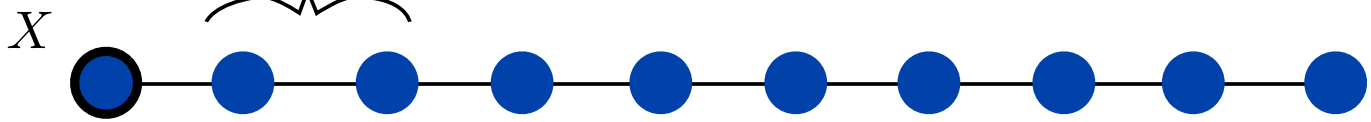
new logical operators

$\bar{X} = [X]_1[Z]_2$
$\bar{Z} = [X]_2[Z]_3$

commute with gauge operators

# There's a Code in My Wire

Z info
localized here



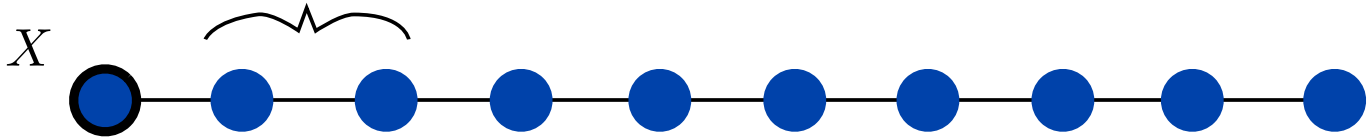

X info
localized here

$\bar{X} = [X]_1[Z]_2$

$\bar{Z} = [X]_2[Z]_3$

but modulo the gauge operators:

Z info
localized here

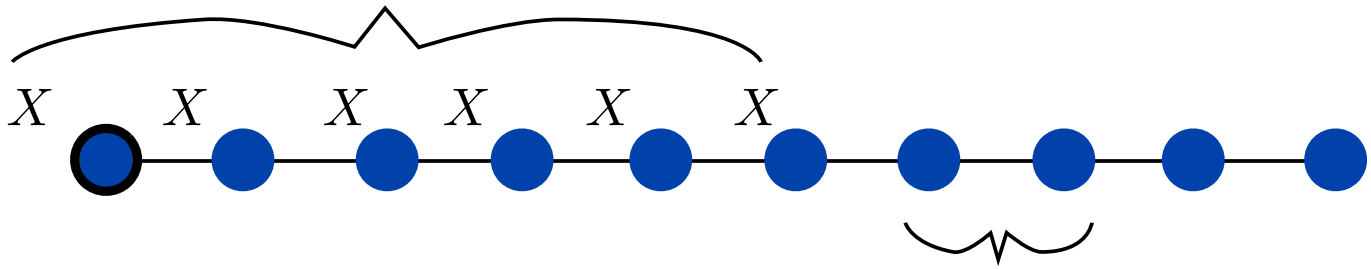



X info
localized here

$\bar{X}\tilde{X}_1 = [Z]_2$

q-info has
propagated down
wire w/ Hadamard

# There's a Code in My Wire




[n, 1, r, 1] subsystem code

Modulo gauge, information can be localized here

$$\bar{X} = [X]_1[X]_3[X]_5[X]_7[Z]_8$$

$$\bar{Z} = [X]_2[X]_4[X]_6[Z]_7$$

Measurement based quantum computing from a subsystem code perspective

# Subsystem Construction

*Take:* set of n-qubit Pauli operators
*These:* generate a group
*This:* group = r qubit Pauli group + abelian group
*Elements:* gauge operators and stabilizer operators

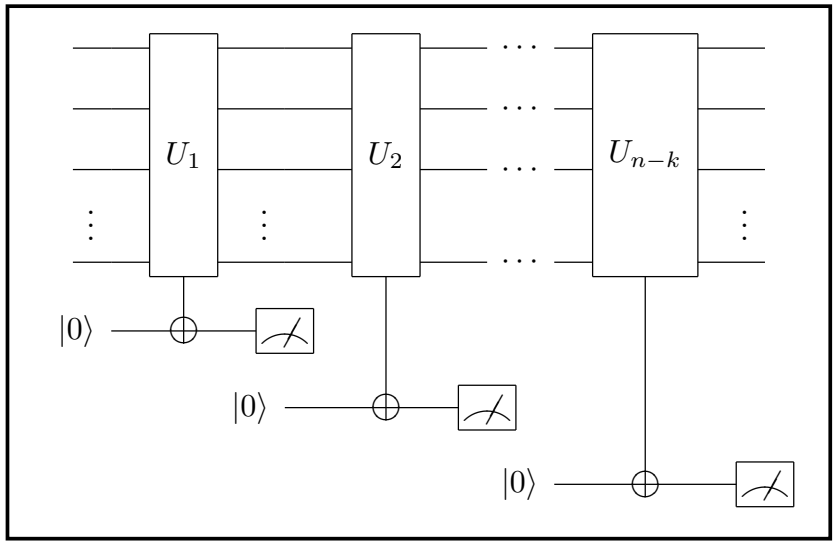
$$\mathcal{T} = \langle T_1, \ldots, T_m \rangle \qquad \mathcal{T} = \mathcal{G} \otimes \mathcal{S}$$

$\mathcal{P}_r$

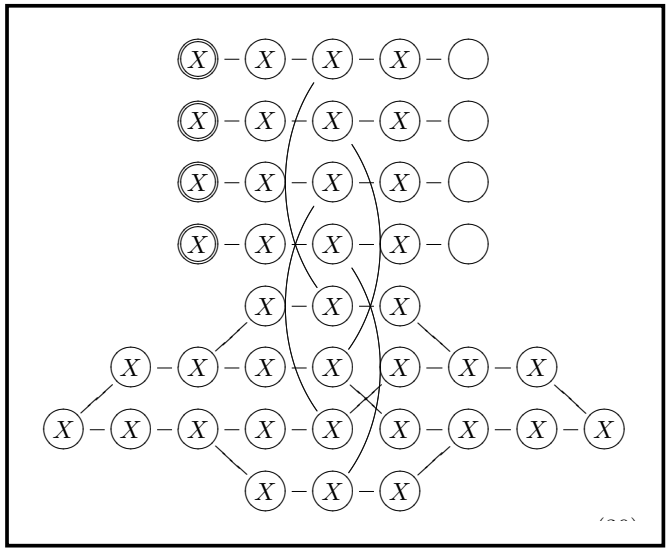
*Left over:* Pauli operators that commute with the group, but not in the group generate logical operators

$$\mathcal{L} = \mathcal{P}_k$$

# Main Idea (WAKE UP)




Circuit for measuring syndrome of a [n,k,d] stabilizer code



MBQC scheme for circuit, convert to subsystem code on N qubits

Claim: RHS code is [N,k,d] code*

* = subject to circuit obeying a FT-like criteria

# Dictionary

circuit to labeled graph

labeled graph describes group

$$\mathcal{T} = \langle T_1, \ldots, T_m \rangle$$

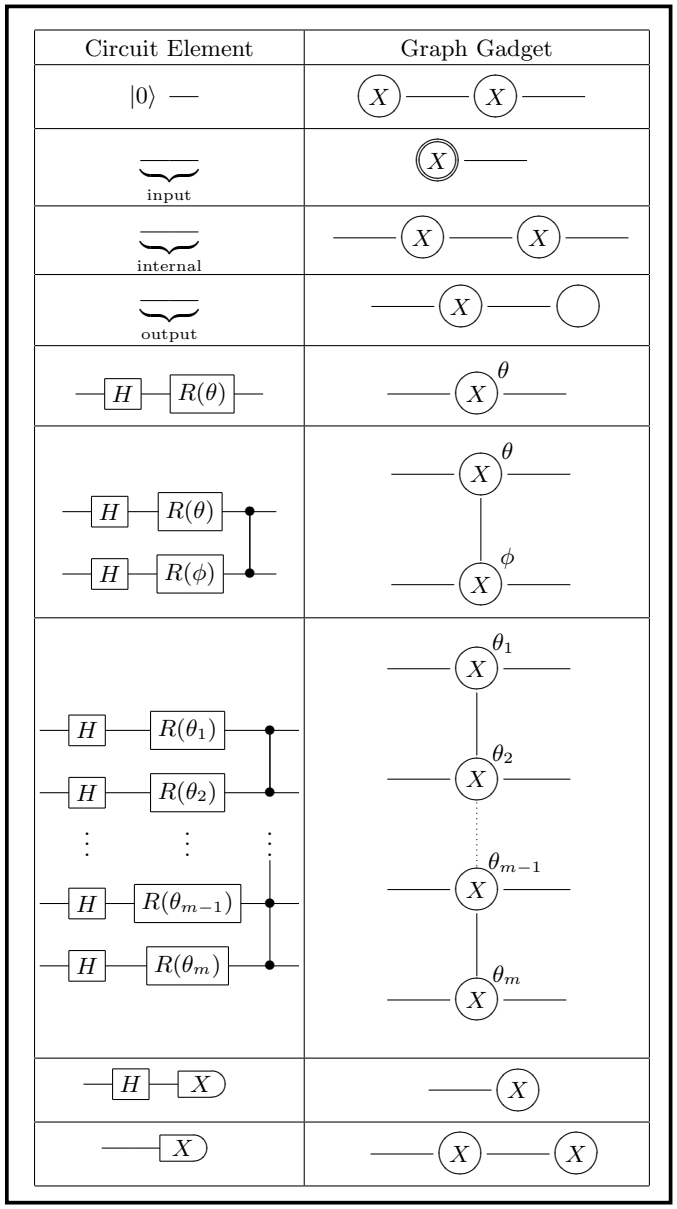
include: vertex operators for non-double framed vertices

$$X_v \prod_{(w,v)\in E} Z_w$$

include: single qubit for each X labeled vertex

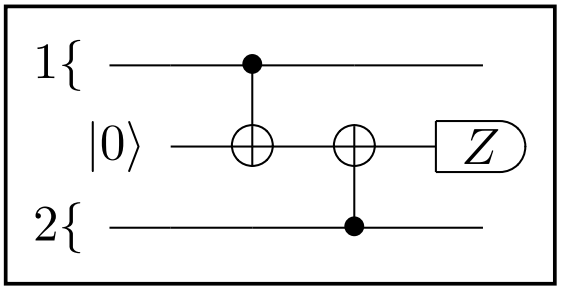
$$\cos\theta_v X_v + \sin\theta_v Y_v$$

$\theta_v = 0$ for no $\theta_v$ label



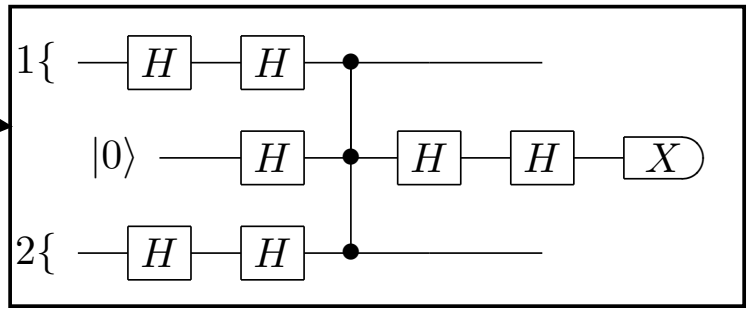
# Circuit, Circuit, Graph

Simplest stabilizer code with stabilizer $S = Z \otimes Z$

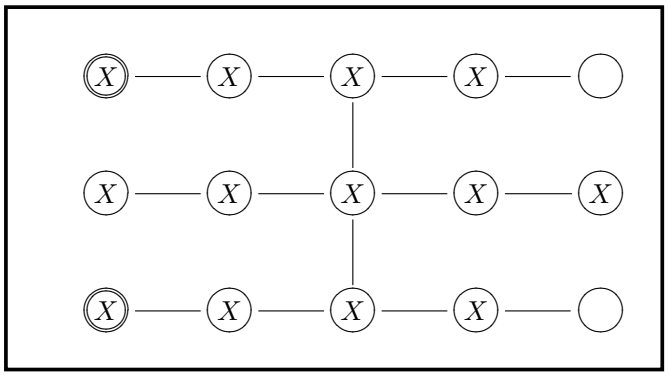



syndrome measuring circuit

convert to
dictionary's gate set



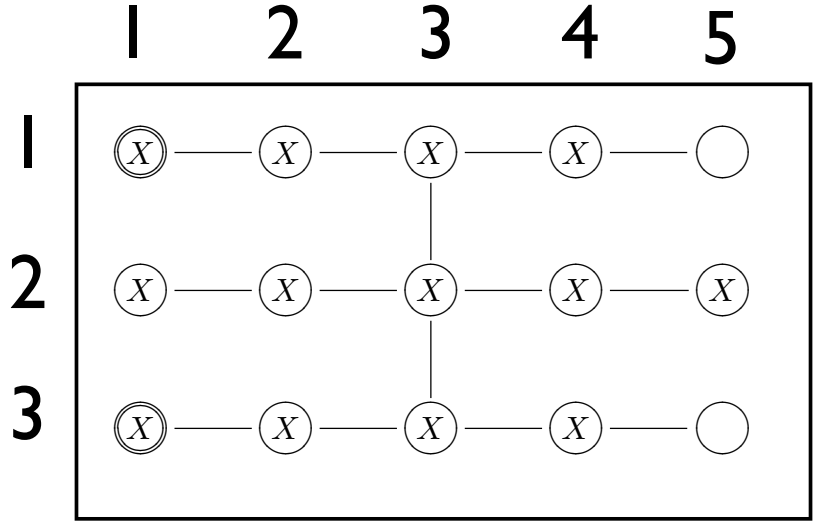

use dictionary to
convert to labeled graph



# Subsystem code

## 12 Gauge Qubits

$$
\begin{aligned}
&\bar{X}_1 = X_{(1,1)}, \bar{Z}_1 = S_{(1,2)} \\
&\bar{X}_2 = X_{(2,1)}, \bar{Z}_2 = S_{(2,2)} \\
&\bar{X}_3 = X_{(a,1)}, \bar{Z}_3 = S_{(a,2)} \\
&\bar{X}_4 = X_{(1,2)}, \bar{Z}_4 = S_{(1,3)} \\
&\bar{X}_5 = X_{(2,2)}, \bar{Z}_5 = S_{(2,3)} \\
&\bar{X}_6 = X_{(a,2)}, \bar{Z}_6 = S_{(a,1)} \\
&\bar{X}_7 = X_{(1,1)}X_{(1,3)}, \bar{Z}_7 = S_{(1,4)} \\
&\bar{X}_8 = X_{(2,1)}X_{(2,3)}, \bar{Z}_8 = S_{(2,4)} \\
&\bar{X}_9 = X_{(1,2)}X_{(1,4)}, \bar{Z}_9 = S_{(1,5)} \\
&\bar{X}_{10} = X_{(2,2)}X_{(2,4)}, \bar{Z}_{10} = S_{(2,5)} \\
&\bar{X}_{11} = X_{(a,4)}, \bar{Z}_{11} = S_{(a,5)} \\
&\bar{X}_{12} = X_{(a,5)}, \bar{Z}_{12} = S_{(a,4)}
\end{aligned}
$$




## 2 Stabilizers

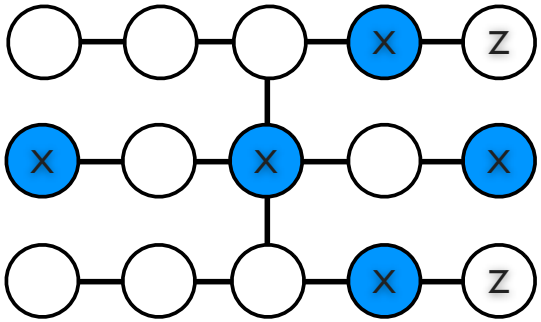
$$
\begin{aligned}
S_1 &= S_{(1,4)}S_{(2,4)}S_{(a,1)}S_{(a,3)}S_{(a,5)} \\
S_2 &= X_{(1,2)}X_{(2,2)}X_{(a,1)}X_{(a,3)}X_{(a,5)}.
\end{aligned}
$$

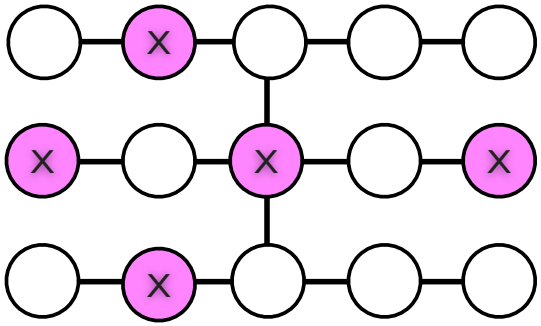
## 1 Logical Qubit

$$
\begin{aligned}
\bar{X}_{L,1} &= X_{(1,1)}X_{(1,3)}X_{(1,5)}X_{(2,1)}X_{(2,3)}X_{(2,5)} \\
\bar{Z}_{L,1} &= X_{(1,2)}X_{(1,4)}Z_{(1,5)}.
\end{aligned}
$$

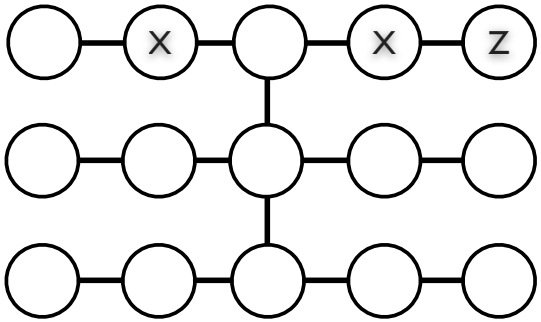
$$
S_{(i,j)} = [X]_{(i,j)} \prod_{(v,(i,j))\in E} [Z]_v
$$

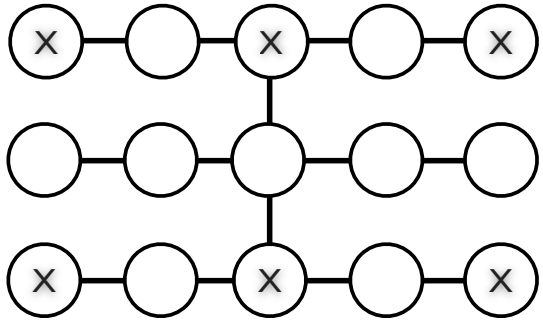
# Code Properties

$S_1 =$ 


$S_2 =$ 

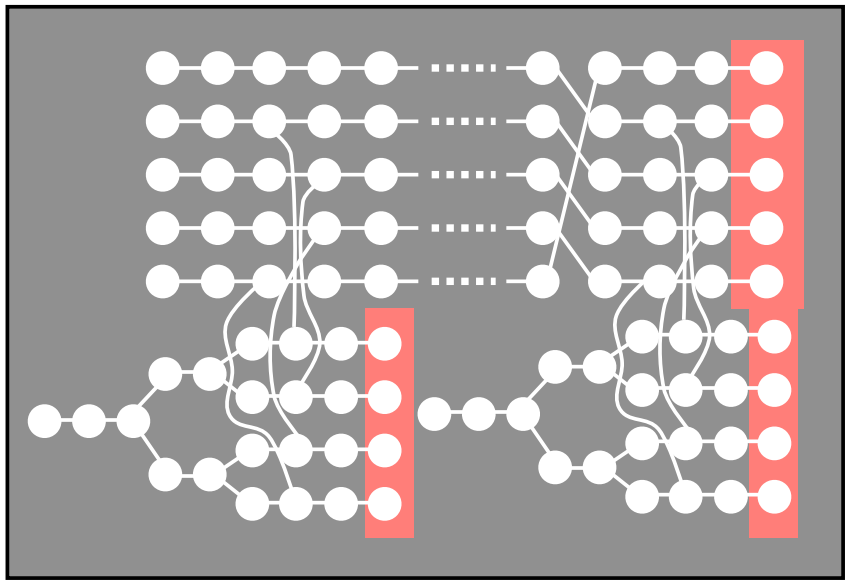

Stabilizers can be measured as product of low weight measurements (blue vertex operators, pink X operators)

$\bar{Z} =$ 

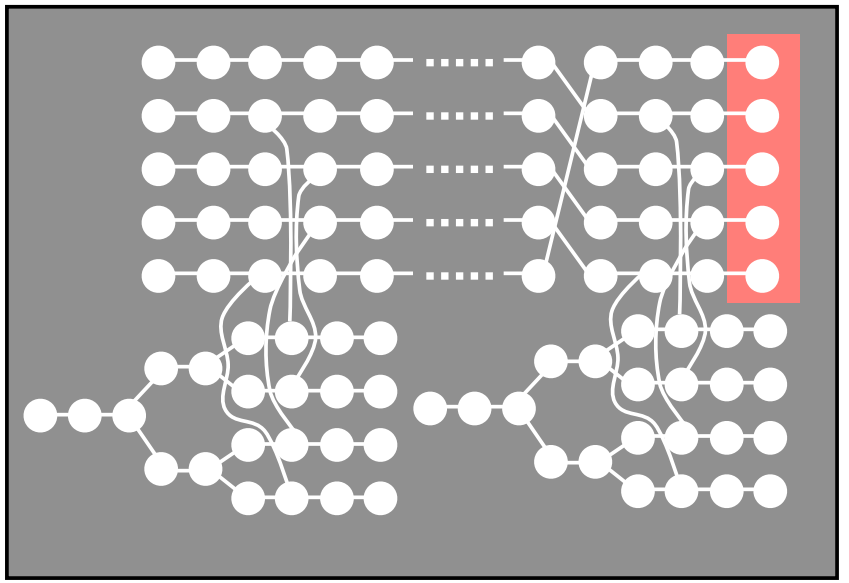

$\bar{X} =$ 


Modulo gauge and stabilizer group logical Z can be made weight 1, but logical X can only be made weight 2

# For Syndrome Circuits

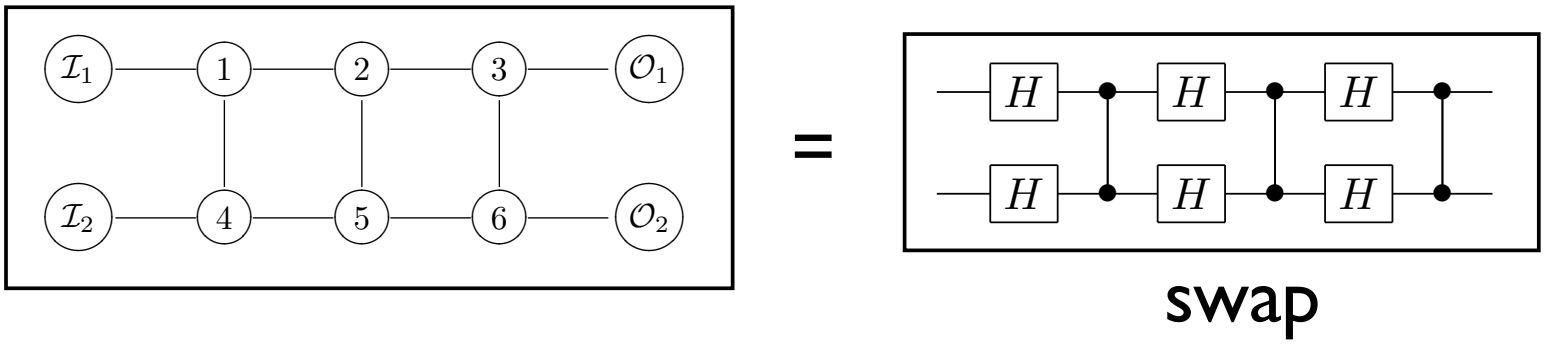


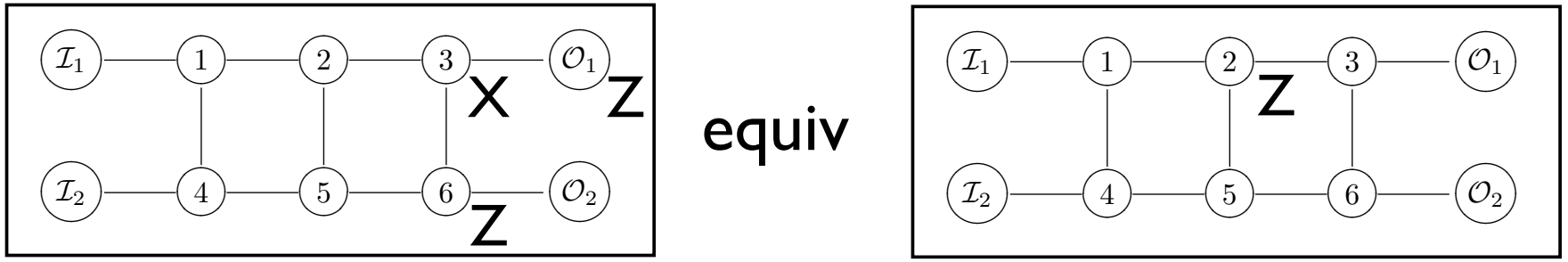
Modulo gauge operators stabilizers are (a) stabilizer of code with ancillas used to measure then (b) ancillas measurement operators

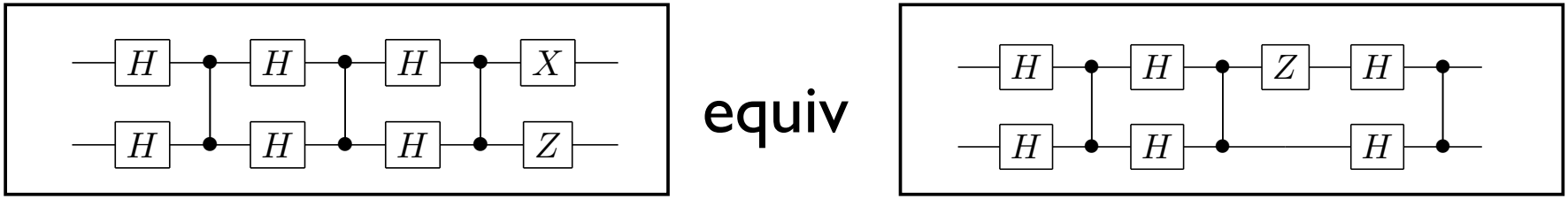


Modulo gauge/stabilizer operators, logical operators are logical operators for code on output qubits

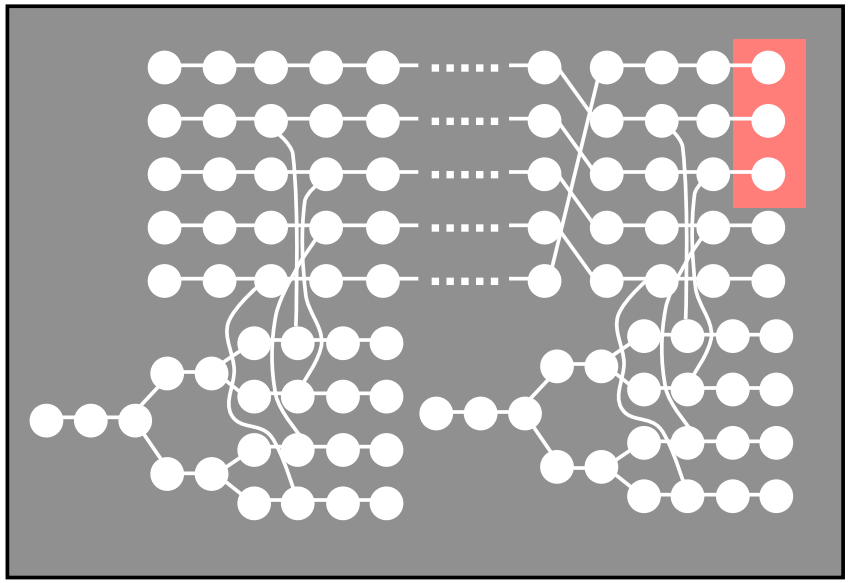
# An Obstacle


$\mathcal{I}_1$
1
2
3
$\mathcal{O}_1$
$\mathcal{I}_2$
4
5
6
$\mathcal{O}_2$
=
H
swap

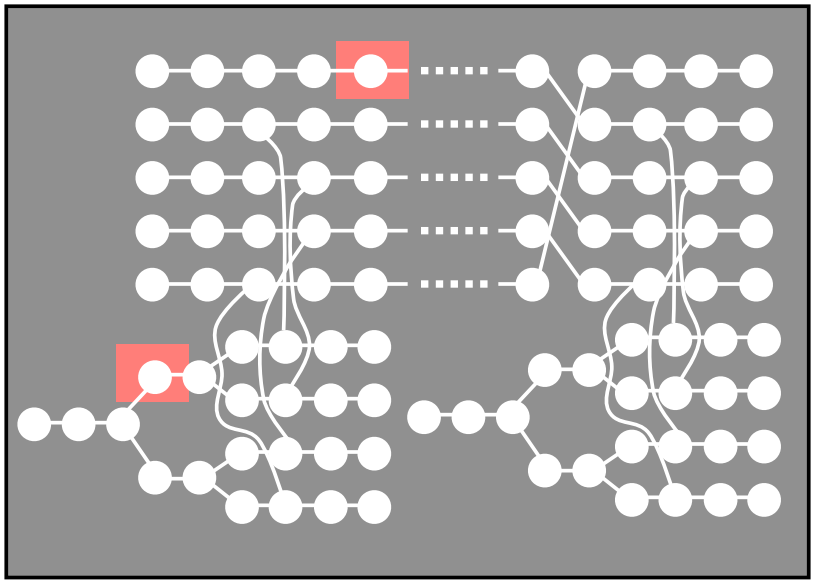

$\mathcal{I}_1$
1
2
3
$\mathcal{O}_1$
X
Z
$\mathcal{I}_2$
4
5
6
$\mathcal{O}_2$
Z
equiv


H
X
Z
equiv

# An Obstacle



weight d logical



weight <d logical

Need to design circuits that do not lower weight of logicals as they are propagated back through code using gauge operators (FT-like criteria)

# Main Result Redux

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] stabilizer subsystem code, $N=O(n^2)$, whose syndrome measurements can be made using c-neighboring measurements (c constant) in $D \geq 2$ dimensions.

Take FT syndrome measuring circuit that is local in D dimensions using dictionaries gate set, convert to labeled graph, and this subsystem code has these properties.

# FAQ

Q: Why fault-tolerant syndrome measuring circuits?

- These maintain the distance of the code, actually we don't need exactly FT circuits, but can get away with a slightly looser criteria.

Q: But aren't fault-tolerant circuits probabilistic?

- Often yes. Use the circuit as if the measurement result turned out in your favor

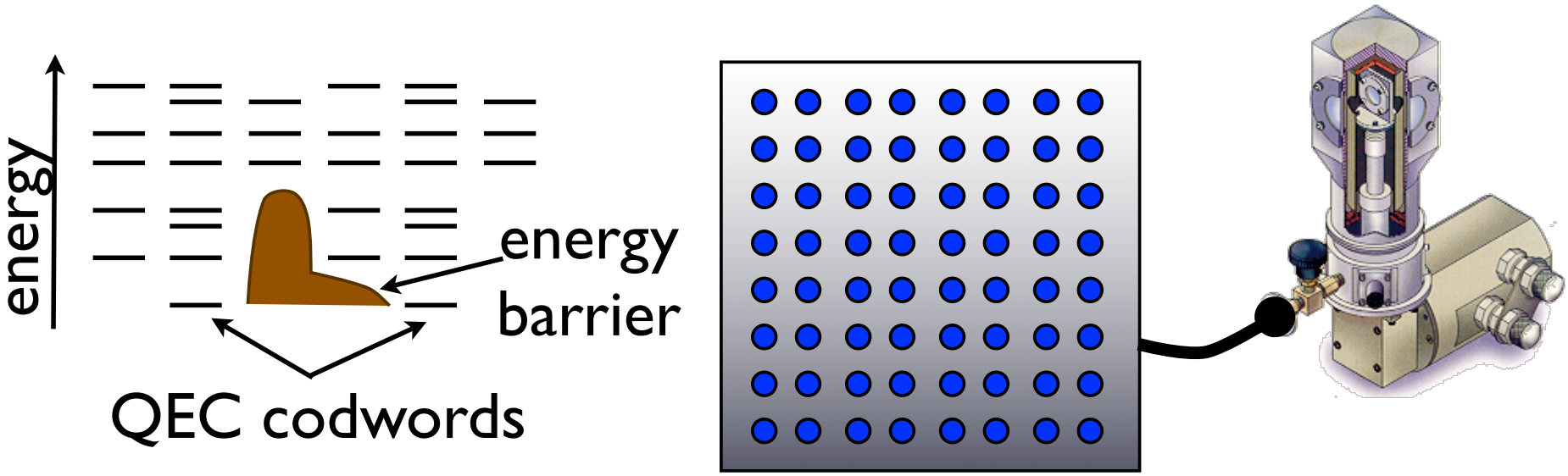
Q: Do these codes have thresholds?

- Unknown, but likely given their relation to fault-tolerant circuits

Q: c-local for what c?

- Can be made 3-local

# QuEST




What happens when we construct a Hamiltonian out of the codes we have constructed above? Self-correcting?

# QW Group

Grads

Gregory Crosswhite (Physics)
Isaac Crosson (Physics)
Lukas Svec (Physics)
Jijiang Yan (Physics)
Kamil Michnicki (Physics)
Kevin Zatloukal (CS)
David Rosenbaum (CS)
Paul Pham (CS)

Undergrads

Zakk Webb (Physics)
Kate Liotta (Physics/CS)
Jonathan Shi (Physics/CS)

Faculty

Dave Bacon
Aram Harrow

http://abstract.cs.washington.edu/~dabacon/